

END395
Operations Research II
Term Project
Spring 2025

In this project, you will work on the multi-period shipment planning problem of a pharmaceutical company that must coordinate shipments between one of its production facilities and a third-party warehouse over a planning horizon of 3 to 7 days. Currently, the company relies on manual planning to determine daily shipments under specific constraints. Your task is to develop a comprehensive mathematical model to optimize this process and improve efficiency.

One of the key logistics challenges pharmaceutical companies face is managing the delivery of finished products to third-party warehouses, which are responsible for fulfilling customer orders. The company in this project encounters a similar issue and seeks an effective shipment planning strategy for transporting finished goods from its production facility to a designated warehouse. The shipment plan must account for several operational constraints detailed below.

Each product has a specific pallet size, and as production continues, the manufactured products are packed into pallets of predetermined sizes. There are two different pallet sizes: 100×120 cm and 80×120 cm. The packed pallets are transported by a fleet of vehicles. The vehicle fleet consists of three types of vehicles: trucks (tır), lorries (kamyon), and vans (kamyonet).

- A truck (tır) can carry up to 22 pallets of size 100×120 cm or up to 33 pallets of size 80×120 cm.
- A lorry (kamyon) can carry up to 12 pallets of size 100×120 cm or up to 18 pallets of size 80×120 cm.
- A van (kamyonet) can carry up to 6 pallets of size 100×120 cm or up to 8 pallets of size 80×120 cm.

In the first part of the project you will assume that any vehicle carry only the pallets of the same size. The vehicle fleet consists of b_k number of vehicles of type $k = 1(\text{truck}), 2(\text{lorry}), 3(\text{kamyonet})$ in the vehicle fleet.

The problem involves vehicle dispatching and packing decisions within the planning horizon T , which consists of a set of consecutive days starting from day 1. Since the transportation time from the production facility to the warehouse is one hour, shipments on day t are assumed to arrive at the destination on the same day. Considering loading, unloading, and transportation times, a vehicle can be assigned to a maximum of three shipments per day.

Let J represent the set of products that are produced and shipped to the warehouse. The pallets of these products to be shipped to the warehouse are represented by the set P . Note that, each pallet $i \in P$ contains n_i products of the same type and may fulfill multiple orders of varying sizes. Based on the production and packing plan, the pallet $i \in P$ becomes available for shipment on day r_i , where $1 \leq r_i \leq T$ and includes products of type $j \in J$. The size of the pallet is indicated by the parameter s_i , where $s_i = 1$ corresponds to 100×120 and $s_i = 2$ corresponds to 80×120 . Let O represent the set

of orders, where p_o represents the number of products required for order $o \in O$, d_o is the due date of order o . Note that, an order's demand may be fulfilled by multiple pallets, potentially delivered on different days.

The loading area at the production facility has limited space for staging pallets before shipment. At the end of each day t , at most q pallets can remain in the waiting area.

The warehouse has sufficient storage capacity to accommodate early deliveries, but storing products before their due dates incurs an earliness penalty. This penalty is represented by a fixed unit cost per day, denoted as s h_o and varies for each order. Despite this penalty, early shipments may still be beneficial if they improve vehicle utilization. However, late deliveries are strictly prohibited. If the available fleet is insufficient for on-time delivery, the company can rent additional vehicles. The shipment cost per trip for owned vehicles is denoted by c_k for vehicle type k , while the trip cost for rented vehicles is denoted by c'_k for each trip of a rented vehicle of type k , where $k = 1(\text{truck}), 2(\text{lorry}), 3(\text{kamyonet})$.

The aim is to determine which pallets should be loaded into each vehicle for each shipment, when shipments should depart, whether rented vehicles should be used, and if so, which type, as well as which pallets should be used to fulfill each order. The goal is to minimize the total cost, which includes both the cost of vehicle usage, whether owned or rented, and the earliness penalty for early deliveries.

A simple example for the problem is given below. In this example, there are three different products, P1, P2, P3. The planning horizon is set to three days and the maximum number of pallets allowed in the waiting area is 15. Additional parameters of the problem are given in Table 1-3. The solution of this example should be submitted as part of the Part I submission.

Pallet ID	Product type	Product amount (n_i)	Size type (s_i)	Pallet type	Release day (r_i)
1	P1	110	1	100×120	1
2	P1	90	1	100×120	1
3	P2	125	1	100×120	2
4	P3	95	2	80×120	2
5	P3	115	1	100×120	1
6	P1	75	2	80×120	3
7	P3	135	1	100×120	3
8	P2	100	1	100×120	1
9	P3	90	1	100×120	2
10	P2	90	1	100×120	3
11	P1	80	2	80×120	2
12	P2	150	2	80×120	1
13	P3	125	1	100×120	3
14	P1	115	1	100×120	2
15	P2	50	2	80×120	1
16	P3	105	1	100×120	1
17	P1	95	2	80×120	2
18	P2	85	2	80×120	2
19	P3	75	2	80×120	1
20	P1	40	2	80×120	2

Table 1: Pallet Data Table

Vehicle	Capacity (100×120)	Capacity (80×120)	Num. of vehicles	Fixed cost (c_k)	Variable cost (c'_k)
Truck	22	33	2	100	150
Lorry	12	18	1	80	120
Van	6	8	1	50	75

Table 2: Vehicle Data Table

Order ID	Product Type	Demand Amount (p_i)	Due date (d_i)	Earliness penalty (h_i)
1	P1	200	2	5
	P2	150	2	5
2	P3	290	3	3
	P1	155	3	3
3	P2	300	3	4
	P3	260	3	4
4	P1	250	2	6
	P2	150	2	6
5	P3	190	1	4

Table 3: Order Data Table

Part I

- Deadline: March, 14, 2024 - Friday 18:30 (**A maximum of three days of latency is permitted, with a penalty of 20 points for each day beyond the deadline.**)
- **Formulate the problem as a mixed integer problem (MIP)** where the objective is to minimize the total cost. The project report should provide a clear description of
 - the decision variables,
 - the objective function, and
 - constraints
- **Time schedule and submission requirements:** The data for three problem instances will be provided. In the project report, provide the following results of each instance in a Results table
 - the objective function value of the best found solution,
 - the solution time (CPU time) of your model, and
 - the percentage gap attained at the end of the run (if you couldn't obtain the optimal solution)
- Project report should be prepared in English and should include all the necessary information specified above. Report format will be announced later and you should prepare your reports in the correct format to get full credit.
- You must submit all your project files (including data sets you used) in a described format. The format will be announced later.
- Submissions and all project related questions should be sent to the course e-mail address. Mails that are sent to the individual e-mail addresses of TA's will not be responded.

Part II

- Deadline: April, 4, 2024 - Friday 18:30 (**A maximum of two days of latency is permitted, with a penalty of 25 points for each day beyond the deadline.**)
- In real life, vehicles of any type can carry different types of pallets together. In this part of the project, you will address this real application in your model. For each vehicle type $k = 1, 2, 3$, there are alternative loading types. For loading type l of vehicle type k , at most α_{kl} pallets of size 100×120 cm and at most β_{kl} pallets of size 80×120 cm can be loaded.
- Data instances for this part will be provided later.
- **For Part II, please explicitly specify the modifications made to the original MIP model submitted in Part I and present a comprehensive formulation of the modified model. Additionally, include detailed explanations for any modifications made for bonuses if achieved.** The project report should provide a clear description of **adjustments or augmentations** in the **decision variables, objective function, and/or constraints**.
- **In the project report, you must provide:**
 - the following results of each instance in a Results table
 - * the objective function value of the best found solution
 - * the solution time (CPU time) of your model
 - * the "% gap" attained at the end of the corresponding run
 - a discussion on the difficulty of the instances based on the results in your Results table.